

Certified primitives for faster unknot recognition

Binary planar tensors, optimal cocycle seeds, and compressed overlap witnesses

A research continuation toward quasi-polynomial complexity

ProveIt research continuation

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Abstract

We continue the performance program in *ProveIt/Topology/UnknotRecognition* from an explicitly pinned repository revision. Three exact improvements are developed, implemented, and tested. First, a two-state planar tensor model computes the full Jones polynomial with at most 2^w frontier states, where w is the crossing-edge cut width of a certified scan order. A combinatorial rotation cochain supplies the correct contribution of every smoothed circle, including diagrams with repeated face walks, loops, and nested circles. The existing planar separator certificate then gives bit time and space $\text{poly}(n)2^{O(\sqrt{n})}$, removing the logarithmic factor in the exponent of the repository's faithful equality-partition backend. Exact integer specialization with balanced-digit recovery and valuation normalization is also provided. This realizes a known square-root-exponential principle in the present implementation; it is not a claim of the first such theorem.

Second, for an integral cocycle in a finite triangulation, we minimize the number of normal discs over all global vertex-potential changes. A sparse integral min-cost flow gives an optimal potential and a matching certificate whose equality with the primal objective proves optimality over both real and integral potentials. The conservative bit bound is $O(T^3(B + \log(T + 1)))$ for T tetrahedra and B -bit local heights. Under the stated geometric hypotheses, an optimum has no nonempty relatively null-homologous component union; a primitive rank-one class therefore gives a connected nonseparating seed. This preserves a property already enjoyed by the initial spanning-tree gauge while reducing its disc count.

Third, an optional compressed cyclic-overlap policy asks for a strict-majority witness before extending its alignment with at most two exact longest common prefix queries. It is complete for the existence of a local shortening move when uncapped, and is much faster on a rigorously specified exponential-word family. It need not find the best greedy move, and the supplied knot corpus shows no demonstrated recognition speedup from this policy.

Proofs, independent checks, controlled measurements, negative results, integration patches, and reproduction scripts accompany the article. None of these improvements establishes a quasi-polynomial worst-case bound for the complete maintained recognizer. We formulate the remaining requirements in terms of compressed geometry, hierarchy resets, state growth, and search completeness, and give a prioritized research agenda with falsifiable targets.

Scope of the result. Theorems in this article certify the stated primitives. Experimental timings measure the supplied instances on one environment. A Jones polynomial equal to one remains an inconclusive result in this implementation. A general quasi-polynomial recognition theorem is *not* claimed.

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1 Research objective and evidence boundary

The objective is to make exact unknot recognition faster while moving toward a defensible quasi-polynomial worst-case analysis. That requires a joint argument about topology, representation size, primitive bit costs, and the number of primitive calls. Speeding up one operation does not supply the missing bounds on the other three. The work below therefore treats an implemented theorem, an observed acceleration, and a proposed global route as different kinds of evidence.

The baseline is the public ProveIt revision

8a95834940cf77cdab1b39571ffc102ca8b6bede

dated 8 October 2026, 15:40:39 UTC. All code changes in the accompanying patch are relative to that revision. The target is `Topology/UnknotRecognition/fast`; the older reports and synthesis documents provide mathematical context and selected independent reference implementations. No remote repository changes are required to inspect or reproduce this package.

1.1 Input size and the asymptotic target

An input knot diagram has n crossings and a validated planar-diagram encoding. Arc labels are normalized during validation; their accidental decimal spelling is not a topological size parameter. We use bit complexity throughout. Writing a bound as polynomial in n permits the usual $O(n \log(n + 2))$ encoding overhead. For a triangulation, T denotes the number of tetrahedra and B the bit length of supplied integral cocycle heights. For a compressed word, the grammar size and the bit length of its represented length are input parameters; the expanded length can be exponential in them.

For an input encoding of length N , quasi-polynomial time means

$$\exp\left((\log(N + 2))^{O(1)}\right).$$

In particular $N^{C \log N}$ is quasi-polynomial, whereas $2^{C\sqrt{N}}$ is a larger asymptotic class. Removing a factor $\log n$ from an exponent $O(\sqrt{n} \log n)$ is a real improvement, but it does not reach the requested target. Conversely, a polynomial operation on a 2^N -letter word is efficient only if the polynomial is in the compressed encoding, not in 2^N .

Three separate resource notions recur:

1. *Mathematical input size*: the diagram, triangulation, grammar, and binary integers specified by the theorem.
2. *Represented intermediate size*: frontier keys, coefficient bits, grammar nodes, attaching maps, or normal coordinates actually stored.
3. *Charged implementation work*: a documented counter or elapsed time allowance. A cap on state count is not a byte-memory cap, and a cap on work does not turn an unfinished search into a negative mathematical answer.

1.2 What the existing implementation already supplies

The maintained recognizer is a portfolio. It validates the input, applies structural and simplification certificates, uses one-sided invariant tests, and can attempt independently replayed group certificates. Complete exact fallbacks include the built-in Khovanov calculation and an optional external normal-surface engine. The previous work already includes separator certificates, exact faithful Potts evaluation, transfer methods for special diagram families, compressed group

words, accelerated repeated automorphisms, whole-donor macros, and a complete local compressed substring primitive.

The relevant baseline guarantees are narrower than a complete recognition bound. A crossing scan with certified edge frontier $O(\sqrt{n})$ is already available. The faithful Potts calculation uses equality partitions, whose Bell-number state count gives $2^{O(w \log(w+2))}$ for w active boundary objects. Its universal full-polynomial bound is thus $\text{poly}(n)2^{O(\sqrt{n} \log n)}$. The compressed word kernel has polynomial local bit bounds, but no general bound on discovery steps or accumulated grammar size follows. The normal-surface adapter can return an exact decision from its external engine, but it does not yet export an independently replayed compressed hierarchy with a quasi-polynomial running-time theorem.

This continuation reuses those working parts. It changes the state space of the full-Jones calculation, optimizes a concrete geometric primitive, and reduces work spent finding a sufficient compressed overlap. The three routes share a principle: a compact representation must preserve the information needed by the next exact operation, and its certificate must describe that operation’s actual conclusion.

1.3 Relationship to the literature

The Kauffman state model and its normalization are classical [5]. Ellenberg, Newman, Sawin, and Shi already proved a polynomial-times- $2^{C\sqrt{n}}$ time and memory bound for the Kauffman bracket using controlled tangle boundaries [4]. General planar Boolean tensor contraction and Jones tensor implementations provide further context [10, 12, 6]. Our contribution here is a self-contained binary model that works with the repository’s existing arbitrary crossing cut order, its exact arithmetic, its input validation, and its resource semantics. The square-root-exponential exponent itself is established prior work.

The geometry route is motivated by Lackenby’s hierarchy construction [7]. It is necessary to distinguish the 2021 announcement of a quasi-polynomial unknot-recognition algorithm [11] from the particular written argument audited here. Section 9 of the July 2026 preprint does not specify all primitive running times; its iteration estimate depends on maximal hierarchy length and maximal pattern-complexity. Section 10 improves the depth control, but the code in this package does not implement all of that geometric machinery. Thus neither the announcement nor the preprint is used as a black-box complexity proof for ProveIt’s recognizer.

Integral optimization through network matrices is also classical. The topological use of total unimodularity has substantial precedent, including optimal homologous cycle algorithms [3]. We claim a precise local-span optimization, certificate, and geometric preservation argument for the implemented cocycle-seed family, without asserting literature-wide priority for network integrality or for connected nonseparating surfaces.

1.4 A current claim about Jones detection and its audited prerequisites

The literature check was made on 8 October 2026. Carmi and Cohen’s June 2026 preprint explicitly claims that the Jones polynomial detects the unknot [2, Theorem 6.9]. The arXiv record inspected on that date lists version 1 only and displays no withdrawal notice. This claim must be acknowledged; it cannot be resolved by repeating older descriptions of the problem’s status. Our theorems and recognition decisions do not assume the claimed detection theorem.

There is a concrete reason for this independence. The pinned repository already contains an audit of a prerequisite consequence of that preprint. Lemma 5.8, combined with the universal realization asserted in Proposition 5.12 and the threshold in Section 5.1.1, implies a uniform bound on the degree in $p = x - x^{-1}$ of every ordinary residual twist state:

$$\deg_p V \leq B_{\text{host}} + 11 + 2.$$

Here the expansion is $V(x) = a(p) + b(p)x$, and the degree means $\max(\deg a, \deg b)$. This implication precedes the low-order anchor vanishing used later in Theorem 5.10; it is not restricted to hosts already known to have trivial Jones polynomial [2, Sections 5.1–5.6].

For clarity, the existing witness can be described without any clasp-machine isotopy. Take the three-crossing host with planar-diagram rows

$$(5, 1, 2, 0), \quad (1, 4, 3, 2), \quad (0, 3, 4, 5).$$

Replace the third row by an odd anti-parallel twist of length L , keeping its exterior ports fixed. Explicitly, the first twist row begins with $(a_0, b_0) = (0, 3)$. At row $i = 0, \dots, L-1$ use (a_i, b_i, c_i, d_i) , where $(c_i, d_i) = (6 + 2i, 7 + 2i)$ for $i < L-1$, $(c_{L-1}, d_{L-1}) = (4, 5)$, and $(a_{i+1}, b_{i+1}) = (d_i, c_i)$. These are the same fixed-exterior diagrams used by the baseline audit.

Their knot Jones polynomials, in $t = x^2$, satisfy the audited skein recurrence

$$V_1(t) = t + t^3 - t^4, \quad V_L(t) = t^2 V_{L-2}(t) + t - t^2 + t^3 - t^4.$$

Consequently the highest term is $-t^{L+3}$. The identity $x^m = F_{m-1}(p) + F_m(p)x$, with

$$F_m(p) = \sum_{k=0}^{\lfloor (m-1)/2 \rfloor} \binom{m-k-1}{k} p^{m-2k-1},$$

shows that the corresponding b -polynomial has leading term $-p^{2L+5}$. All lower t -powers contribute smaller p -degrees. Thus the degree is $2L + 5$; for the mirror it is $2L + 6$.

The host has $B_{\text{host}} = 3 \cdot 3 + 1 - 1 = 9$, so the proposed uniform residual cap is 22. The baseline's $L = 9$ witness has degree 23. The present package independently recomputes its complete Jones polynomial with Regina 7.4.1, and recomputes the formal expansion using the displayed binomial formula rather than the baseline's coefficient recurrence. It also checks $L = 23$: the degree is 51, while $L > 22$ even satisfies the long-twist inequality needed by the later descent. The mirror degrees are 24 and 52, respectively. All four polynomial computations agree coefficient by coefficient with all three tensor arithmetic modes.

The original audit additionally records local oriented-smoothing reductions and Reidemeister-II anchor cancellations. Our new independent computation replicates its polynomial obstruction. It neither constructs a nontrivial knot with Jones polynomial one nor identifies a unique erroneous clasp-machine step. It shows that the stated universal geometric realization and uniform residual-degree consequence cannot both apply to these explicit ordinary twists. Accordingly, a Jones polynomial different from one gives the usual nontriviality obstruction, whereas identity remains inconclusive in this implementation. The source and complete raw outputs are `reproduce/audit_regina_jones.py` and `results/regina_jones_20261008.json`.

1.5 The principal results and their scope

Primitive	Proved improvement	Remaining limitation
Full Jones polynomial	Binary frontier states; exact $\text{poly}(n)2^{O(\sqrt{n})}$ bit bound with a certified separator order	Subexponential rather than quasi-polynomial; identity is not accepted as an unknot certificate by this implementation.
Cocycle normal seed	Exact minimum disc count within a fixed vertex-potential family; polynomial binary cost and a short dual certificate	Restricted family; no incompressibility, genus-minimality, or hierarchy-reset bound.
Compressed overlap	A sufficient strict-majority witness followed by at most two LCP extensions; exact local-existence completeness when uncapped	May choose a worse greedy move; preceding stages and total search remain possible bottlenecks.

Table 1: Theorem scope. Each result is used only for the conclusion it proves.

The paper’s algorithmic claims are proved below and checked against independent oracles where feasible. Those checks increase confidence in the implementation; finite tests do not establish the asymptotic theorems. Conversely, a correct proof of an algorithm does not excuse a mismatch between the mathematical algorithm and the code. The package therefore retains full inputs, exact outputs, proof traces where available, source hashes, and censored observations.

2 Exact full Jones computation with binary planar tensors

The extra logarithm in the faithful partition backend comes from remembering an equality relation among frontier objects. A binary tensor model can instead remember one orientation bit per frontier edge. The difficulty is the loop factor: summing the two orientations of a smoothed circle must produce the Kauffman factor, independently of its shape or nesting. A combinatorial rotation cochain supplies this information without drawing coordinates or floating-point angles.

2.1 A checked rotation cochain

Assume first that $n > 0$. The validated one-component classical diagram gives a connected four-valent graph cellularly embedded in the sphere. Multiple edges, loops, and repeated vertices in a face boundary are allowed. Its $4n$ darts are numbered by crossing and counterclockwise port. Let α reverse an edge and let ρ advance one port at a crossing. The face walks are the cycles of $\rho\alpha$, in the convention used by the implementation. There are $F = n + 2$ faces and $2n$ edges. A distinguished face o is regarded as the outer face.

Assign an integer bend $\beta(d)$ to each dart, with $\beta(\alpha d) = -\beta(d)$. The local corner of a positive face-boundary walk contributes one quarter turn. We require

$$\sum_{d \in f} \beta(d) = b_f, \quad b_f = \begin{cases} 4 - |f|, & f \neq o, \\ -4 - |o|, & f = o. \end{cases} \quad (1)$$

Thus bounded faces have total turn 4 and the outer face has total turn -4 . The word “positive” here denotes the selected permutation convention; no additional claim that a face lies on the left of its traversal is needed.

The demand sum is zero:

$$\sum_f b_f = 4(F - 1) - 4 - 4n = 0.$$

Root a spanning tree of the dual graph at o , and send each subtree's total demand through its parent edge. Put zero bend on non-tree dual edges. This constructs an integral antisymmetric solution. The implementation checks every antisymmetry equation and every face sum after routing; it does not accept the routing procedure itself as the check.

Lemma 2.1 (A polynomial bend envelope). *The routed cochain can be chosen to satisfy*

$$\max_d |\beta(d)| \leq 3(n+1), \quad \sum_{\{d, \alpha d\}} |\beta(d)| \leq 3(n+1)^2.$$

Proof. Every face is nonempty. A positive demand is at most 3, and only the $n+1$ bounded faces can have positive demand. Since total demand is zero, the absolute demand of any set of faces is at most total positive demand, which is at most $3(n+1)$. The dual spanning tree has $F-1 = n+1$ edges. The routed value on each is a subtree demand sum, giving both bounds. $\square$

No metric embedding is required. In particular, long arithmetic coordinates from a planar straight-line drawing never enter this calculation.

2.2 Why every smoothed circle has rotation four

A smoothing at a crossing joins adjacent ports. Write $\tau(a, b) = +1$ when $b - a \equiv 1 \pmod{4}$ and $\tau(a, b) = -1$ when $b - a \equiv -1 \pmod{4}$. Traversing a smoothing from incoming port a to outgoing port b contributes $\tau(a, b)$. An edge traversal contributes the corresponding signed bend. We must prove that the total is always $+4$ or -4 on an oriented smoothed circle.

Lemma 2.2 (Rotation of a smoothing component). *For any choice of all n smoothings, every resulting circle has total combinatorial rotation $+4$ in one orientation and -4 in the other.*

Proof. Work with abstract face disks rather than assuming that each face boundary is a simple polygon in the projection graph. This retains the correct local sectors at a loop or a repeated vertex. A chosen smoothing creates a band between the two uncapped opposite face sectors. On the boundary of the region receiving the band, two old $+1$ corners become two -1 turns. Its total boundary rotation decreases by 4, exactly as its Euler characteristic decreases by 1 when a band is attached.

Initially each bounded face disk has rotation 4 by (1). Consequently every bounded complementary region R of the smoothed circles has total boundary rotation $4\chi(R)$. This argument permits bands within one region and between distinct regions; it depends only on the Euler change under a band attachment.

An innermost smoothed circle bounds a disk and therefore has rotation 4 in its positive boundary direction. Now proceed outward through the nesting of circles. A bounded complementary region with h holes has $\chi(R) = 1 - h$. Each already treated inner boundary contributes -4 with the region's boundary direction. Therefore its outer boundary has rotation $4(1 - h) + 4h = 4$. This proves the assertion for all circles. Reversing direction negates every corner turn and bend, giving -4 . $\square$

This proof addresses precisely the cases that a naive appeal to a polygonal turning-number picture can overlook: repeated face walks, self-loops, and nested components. It also explains why assigning a formal loop weight at an arbitrary frontier connectivity is unnecessary in the binary model.

2.3 Local tensors and normalization

Use the Kauffman bracket convention in which a state with c circles has loop factor δ^c , with

$$\delta = -A^2 - A^{-2}.$$

The normalized Jones polynomial divides by one loop factor. Make the formal substitution

$$A = iz^2, \quad \delta = z^4 + z^{-4}.$$

Factor out one i per crossing. Smoothing $s = 0$ then has weight z^2 , and smoothing $s = 1$ has weight $-z^{-2}$. All remaining arithmetic is in $\mathbb{Z}[z, z^{-1}]$.

At a crossing let $\varepsilon_j \in \{+1, -1\}$ denote the orientation of port j , positive when pointing out of the crossing. Edge contraction identifies opposite port signs. A smoothing joins opposite signs, so a nonzero crossing tensor has exactly two positive signs and two negative signs: at most six assignments. The pairings are

$$\mathcal{P}_0 = \{(0, 1), (2, 3)\}, \quad \mathcal{P}_1 = \{(0, 3), (1, 2)\}.$$

For a compatible pair (a, b) the directed corner contribution is $-\varepsilon_a \tau(a, b)$, because $\varepsilon_a = -1$ means that the traversal enters at a . Absorb an edge bend at the smaller-numbered dart only. If $E_v(\varepsilon)$ is the sum of the absorbed signed bends at crossing v , the local tensor entry is

$$T_v(\varepsilon) = \sum_{\substack{s \in \{0, 1\} \\ \varepsilon_a = -\varepsilon_b \ ((a, b) \in \mathcal{P}_s)}} (-1)^s z^{2-4s+E_v(\varepsilon)-\sum_{(a, b) \in \mathcal{P}_s} \varepsilon_a \tau(a, b)}. \quad (2)$$

Each entry has at most two signed monomials. A loop edge at one crossing enforces its two endpoint signs locally and never creates a frontier edge.

For clarity, the entries before multiplication by z^{E_v} are as follows. The asymmetry comes from the fixed turning convention; the closed invariant is independent of it.

$(+, +, -, -)$	$(+, -, +, -)$	$(-, +, +, -)$	$(+, -, -, +)$	$(-, +, -, +)$	$(-, -, +, +)$
$-z^{-2}$	0	z^2	z^2	$z^4 - z^{-4}$	$-z^{-2}$

The upper bound six is deliberately conservative and remains valid when local cancellations or loop constraints remove assignments.

Let $F_D(z)$ be the contraction of these tensors. Expanding each local smoothing choice and then each compatible orientation orients every smoothed circle independently. Lemma 2.2 makes its two orientation weights sum to $z^4 + z^{-4}$. Therefore

$$\langle D \rangle|_{A=iz^2} = i^n F_D(z). \quad (3)$$

If $w(D)$ is the writhe, the usual oriented normalization gives

$$V_D(z^{-8}) = (-1)^{(n-w(D))/2} z^{-6w(D)} \frac{F_D(z)}{z^4 + z^{-4}}. \quad (4)$$

The exponent $(n - w(D))/2$ is integral. The Laurent implementation performs an exact division with a checked zero remainder, applies the sign and shift, and checks that every remaining z -exponent is divisible by eight. It publishes the coefficient dictionary in variable $t = z^{-8}$. The empty crossing diagram is treated explicitly as the one-component unknot.

Proposition 2.3 (Conservative output bounds). *For an n -crossing knot diagram,*

$$V_D(t) = \sum_e c_e t^e, \quad |e| \leq 2n, \quad \sum_e |c_e| \leq 4^n.$$

Proof. In the state sum after division by one loop factor, a state contributes a monomial times δ^{c-1} . There are 2^n states and at most $n + 1$ circles in a state of a connected n -crossing projection. Its coefficient mass is at most $2^{c-1} \leq 2^n$, so the sum has mass at most 4^n . The smoothing monomial has A -degree between $-n$ and n ; the loop factor adds at most $2n$ in absolute degree, and the writhe correction adds at most $3n$. Thus the normalized A -degree is at most $6n$ in absolute value. The standard knot normalization has integral $t = A^{-4}$ exponents, which gives the more than sufficient bound $|e| \leq 2n$. $\square$

The code also verifies $V_D(1) = 1$ and these bounds. These are useful error checks, not a substitute for the state-sum proof or an independent polynomial oracle. A wrong polynomial can satisfy all three.

2.4 Frontier contraction and the square-root bound

Fix an ordering of the crossings. After a prefix has been processed, let ∂S be the edges with exactly one processed endpoint. A state is one bit for each edge of ∂S , with the bit referenced to its canonical dart. The coefficient for that bit assignment is the sum of all compatible tensor products in the processed part. The update joins the current crossing to every compatible assignment, forgets closed edges, and introduces the new frontier edges. It sums equal output assignments. There is no remembered partition and no assumption that the processed part is a disk.

Let w be the maximum number of crossing edges in a prefix cut. There are at most 2^w keys and at most six local transitions per incoming key. Indexing the bits costs only polynomial overhead. At any intermediate stage the formal exponent envelope is

$$|e| \leq B_n := 4n + 3(n + 1)^2. \quad (5)$$

Indeed a crossing contributes a smoothing exponent of absolute value at most two and two corner turns of total absolute value at most two; the absorbed bends contribute at most the cochain's total absolute bend. Expanding at most six assignments and two monomials per crossing bounds the total coefficient mass of any partial sum by 12^n . A coefficient therefore has $O(n)$ bits, and there are $O(n^2)$ possible Laurent exponents per key.

Theorem 2.4 (Binary frontier full-Jones bound). *Given a validated n -crossing knot diagram and an order of crossing-edge frontier width w , the full Jones polynomial can be computed exactly in $\text{poly}(n)2^w$ bit time and space. With the repository's checked planar separator order this becomes*

$$\text{poly}(n)2^{O(\sqrt{n})}.$$

Proof. The invariant of the update is the partial contraction just described; induction on the prefix proves it. At the end the frontier is empty, so (3) and (4) give the result. The state count and polynomial arithmetic bounds prove the first claim.

For completeness, a bounded-degree planar graph has recursively balanced separators of size $O(\sqrt{m})$ on an m -vertex subproblem [9]. List each child subproblem before its separator, as in the maintained constructor. At a prefix inside a child, a crossing edge leaving the prefix is incident to one of the ancestor separators or the current leaf. Since degree is four, the frontier is bounded by four times the sum of the separator sizes on that recursion path. Balance gives a geometric series $\sqrt{n} + \sqrt{\theta n} + \sqrt{\theta^2 n} + \dots = O(\sqrt{n})$ for some fixed $\theta < 1$. The maintained separator constructor supplies an explicit partition hierarchy; its verifier checks separation, balance, size, the flattened order, and the resulting width. The tensor backend constructs and checks this witness before relying on its guarantee. $\square$

The implementation can compare this order with an ordinary scan order and use the narrower one; the certified witness still bounds the chosen width. An explicit `certified=False` call is a useful ordering ablation, but it has no universal square-root width promise. Supplying a bad order does not invalidate the invariant, only its performance.

2.5 Exact specialization and balanced-digit recovery

Python sparse Laurent dictionaries carry appreciable overhead. We can evaluate the tensors at one sufficiently large exact integer and reconstruct the entire final polynomial, retaining the same frontier state count. This is a coefficient-bounded Kronecker substitution, not a numerical sample used as probabilistic evidence.

Choose

$$k = \max\left(1, \left\lceil \frac{2n+2}{8} \right\rceil\right), \quad z = 2^k, \quad x = z^8 \geq 4 \cdot 4^n.$$

For a first implementation, subtract each local tensor's minimum exponent g_v and contract its integral value. If $g = \sum_v g_v$, the final integer is $Z = z^{-g} F_D(z)$. Equation (4) yields

$$P(x) := x^{2n} V_D(x^{-1}) = \frac{(-1)^{(n-w)/2} z^{16n+4-6w+g} Z}{z^8 + 1}. \quad (6)$$

A negative power of z is moved into the denominator. Exact integer division must have zero remainder. The polynomial P has degree at most $4n$, and each coefficient has absolute value at most $4^n < x/2$.

Lemma 2.5 (Balanced recovery). *An integer of the form $P(x) = \sum_{j=0}^{4n} a_j x^j$, with $|a_j| < x/2$, determines all the a_j uniquely. Iteratively choose the residue in $[-x/2, x/2)$ and divide the remaining difference by x .*

Proof. The constant coefficient is the unique allowed representative of the integer modulo x . Subtracting it and dividing by x leaves the same problem with one fewer coefficient. Induction proves uniqueness and the algorithm. The recovered coefficient at index j is the coefficient of t^{2n-j} in $V_D(t)$. $\square$

The intermediate specialized tensor values need not encode their formal Laurent polynomials injectively. Distinct partial polynomials can specialize to the same value, and a nonzero partial polynomial can specialize to zero. Evaluation is a ring homomorphism, so every subsequent addition and multiplication still gives the correct final value. Faithfulness is needed only for final recovery, where the proved degree and coefficient bounds apply. This distinction permits much smaller intermediate keys and avoids an unjustified assumption about partial coefficient growth.

The implementation includes an exact identity shortcut: compare the contracted value to the value implied by $V_D = 1$ before normalization and digit extraction when its monomial exponent permits that comparison. The same faithful encoding proves this equality is a whole-polynomial identity test. It does not make the invariant a complete unknot certificate.

2.6 Valuation normalization and gauge covariance

The global-shift version can spend most of its time carrying powers of z that merely encode monomial offsets. Store a nonzero specialized value in the canonical form

$$z^v m, \quad v \in \mathbb{Z}, \quad m \in \mathbb{Z} \setminus \{0\}, \quad z \nmid m.$$

Zero has a separate representation. Multiplying by a signed local monomial adds to v and changes the sign of m . To add $z^{v_1} m_1$ and $z^{v_2} m_2$, put $r = \min(v_1, v_2)$ and form

$$a = m_1 z^{v_1-r} + m_2 z^{v_2-r}.$$

If $a \neq 0$, remove the largest power z^j dividing a and return $(r + j, a/z^j)$. With $z = 2^k$, this uses the number of trailing zero bits of $|a|$, divided by k . It is exact even when carries create additional factors of z . These are valuations of the evaluated rational number, not formal lowest exponents of an unexpanded polynomial.

At the final state let $F_D(z) = z^v m$. Equation (6) applies with $g = v$ and $Z = m$, and the same balanced decoding recovers the full polynomial. From (5) and the 12^n mass bound,

$$-B_n \leq v \leq B_n + \log_z(12^n), \quad \text{bits}(m) \leq 2kB_n + n \log_2 12 + O(1) = O(n^3).$$

Thus arbitrary-precision integer work remains polynomial per transition. The bound is deliberately conservative; the experiments record actual mantissa sizes.

Proposition 2.6 (Gauge covariance of partial contractions). *Fix the same outer face. If β and β' both satisfy (1), their difference is an integral vertex coboundary on the projection graph. For each fixed frontier assignment, the two partial contraction coefficients differ by a monomial z^q depending only on that assignment and the processed crossing set. Closed contractions agree. Consequently the normalized mantissas and their bit lengths agree.*

Proof. The difference has zero period on every facial cycle. Since the embedding is spherical, facial cycles generate the integral cycle space, so integrating along a primal spanning tree yields an integral potential h with $\beta' - \beta = \delta h$. In a tensor product, the resulting exponent is a sum of signed edge potential differences. At each internal crossing, the two-in/two-out condition cancels the coefficient of h . Only boundary terms remain; the convention for absorbing a bend at one endpoint fixes which boundary term is used. All internal assignments with the same frontier signs therefore receive the same monomial multiplier. It factors through every partial sum. At a closed frontier the multiplier is one. Canonical valuation normalization removes precisely such a multiplier from the mantissa. $\square$

This also gives a useful kernel-level robustness statement. An arbitrary valid same-outer cochain with L -bit bends can have enormous numerical values. Its boundary gauge offsets need only $O(L + \log(n + 2))$ bits; the mantissas and the differences used to align terms are those of the bounded canonical cochain. A valuation implementation therefore has $\text{poly}(n, L)2^w$ cost on such data. The public constructor currently creates the bounded canonical cochain; accepting externally supplied gauges is not a new public interface in this patch.

Gauge covariance alone does *not* prove that every intermediate formal Laurent span is $O(n)$. It removes a common monomial offset, not the span between genuinely different exponents. We retain the proved $O(n^2)$ envelope for the complexity theorem and leave sharper span bounds as a separate research problem.

2.7 Recognition semantics and limits

A completed polynomial different from one proves knottedness. A completed identity polynomial is retained as evidence and the recognizer continues to its other stages. An exhausted state, transition, or global time allowance publishes no partial polynomial as an invariant. This distinction is checked both through the standalone Jones command and through the public recognizer.

The patch introduces `jones_backend="tensor"` and the CLI choice `-backend tensor`. The optional public dispatch uses valuation-normalized integer arithmetic. The low-level `tensor_jones` function defaults to the Laurent implementation as a readable reference and supports the two integer modes explicitly. The existing default Jones backend is unchanged. The square-root theorem concerns uncapped complete invariant computation; capped queries are partial decision attempts and may be inconclusive.

3 Certified minimization of cocycle normal seeds

3.1 The precise family being optimized

Let $\mathcal{T}$ be a finite triangulation of a compact orientable 3-manifold M . Generalized triangulations with distinct faces of the same tetrahedron identified are allowed, provided they genuinely triangulate the stated manifold. Ideal vertices must first be truncated. Choose an orientation for every global edge, and let c be an integral simplicial 1-cocycle. In tetrahedron t , write $v_{t,i}$ for the global vertex represented by local corner $i \in \{0, 1, 2, 3\}$. There are integral local heights $h_{t,i}$, unique up to an additive constant for each tetrahedron, satisfying

$$h_{t,j} - h_{t,i} = c([i, j]_t).$$

The cocycle equation on triangular faces guarantees their existence. On a paired face the two lists of heights differ by a constant integer.

For an integral global vertex potential f , put

$$c_f = c + \delta f, \quad a_{t,i} = h_{t,i} + f(v_{t,i}).$$

The half-integral levels of the affine local height functions fit together to give a cooriented normal surface $S(c_f)$. They fit because paired-face height functions differ by an integer and we use all levels in $\mathbb{Z} + \frac{1}{2}$. Its signed intersection with each oriented edge is $c_f(e)$; in particular it represents the relative homology class Poincaré dual to $[c]$.

Order the four local corners so that $a_0 \leq a_1 \leq a_2 \leq a_3$. The three successive gaps count respectively triangles at the lowest corner, quadrilaterals separating the lowest two corners from the highest two, and triangles at the highest corner:

$$a_1 - a_0, \quad a_2 - a_1, \quad a_3 - a_2.$$

Zero gaps cause no ambiguity when heights coincide. Thus there is at most one nonzero quadrilateral type, and the number of discs in tetrahedron t is exactly its height span. Our objective is consequently

$$D(f) = \sum_{t=1}^T \left(\max_i a_{t,i} - \min_i a_{t,i} \right). \quad (\text{CS1})$$

This is the number of normal discs, not the Euler characteristic, genus, Thurston norm, or pattern-complexity. All coordinates are written in binary; the construction takes three gap operations per tetrahedron.

3.2 A sparse network and a short dual certificate

Theorem 3.1 (Integral span minimization). *Given T lists of four integer heights and four global vertex labels per list, the minimum of (CS1) over integral potentials can be computed in time polynomial in their binary input size. The same minimum is obtained over real potentials. There is an optimality certificate consisting of an integral potential and T matched corner pairs. It can be checked with $O(T)$ integer arithmetic operations after indexing the vertex labels.*

Proof. Introduce free variables L_t, U_t and f_v and minimize

$$\sum_t (U_t - L_t)$$

subject to

$$U_t - f_{v_{t,i}} \geq h_{t,i}, \quad f_{v_{t,i}} - L_t \geq -h_{t,i} \quad (t = 1, \dots, T; \ i = 0, 1, 2, 3). \quad (\text{CS2})$$

For fixed f , minimizing over L, U recovers (CS1) exactly. Each constraint is a difference between two variables, so its coefficient matrix is a directed incidence matrix, up to transposition. In particular, this is an integral network linear program. The following explicit dual description also yields a constructive proof of integrality.

Make nodes ℓ_t , u_t , and one node for each global vertex v . Add four arcs $\ell_t \rightarrow v_{t,i}$ of reward $-h_{t,i}$ and four arcs $v_{t,i} \rightarrow u_t$ of reward $h_{t,i}$. Each ℓ_t has supply 1, each u_t has demand 1, and the vertex nodes have balance zero. Maximize total reward. The network is feasible: send a unit from each ℓ_t through any one of its corners to u_t . Its underlying directed graph has three layers, so a feasible flow has no directed circulation, and every arc carries between zero and one unit. Network integrality gives an integral optimum.

An integral flow selects one corner (t, i) leaving each lower node and one corner (s, j) entering each upper node. At a global vertex, match the incoming selected corners arbitrarily to the outgoing selected corners. This produces T rows (t, i, s, j) such that sources t and targets s each occur once, and $v_{t,i} = v_{s,j}$ in every row. The dual value is

$$Q = \sum_{(t,i,s,j)} (h_{s,j} - h_{t,i}). \quad (\text{CS3})$$

For any real potential and any such row, the common vertex potential cancels:

$$\max_k a_{s,k} - \min_k a_{t,k} \geq a_{s,j} - a_{t,i} = h_{s,j} - h_{t,i}.$$

Summing over the permutations of lower and upper nodes gives $D(f) \geq Q$. Therefore a potential and a matching with $D(f) = Q$ certify optimality even against all real potentials. This verifier needs no flow solver.

For completeness, an optimal flow gives an integral potential achieving equality. In its residual graph retain every forward arc with its original reward, and retain the reverse arc with negated reward whenever the original flow is positive. Optimality says that there is no positive-reward directed cycle. Add a universal source with zero-reward arcs and take longest-path distances x . These are finite integers. For each residual arc $p \rightarrow q$ with reward r , they satisfy $x_q - x_p \geq r$. On each positive-flow arc the reverse inequality is also present, giving equality. Set $f_v = x_v$, $L_t = x_{\ell_t}$, and $U_t = x_{u_t}$. These satisfy (CS2), and multiplying the equalities on used arcs by their flows and summing gives

$$\sum_t (U_t - L_t) = Q.$$

The actual span objective of this f is at most the left side, while weak duality gives the opposite inequality $D(f) \geq Q$. Hence equality holds. $\square$

Implementation and binary complexity. The implementation negates rewards to obtain a min-cost problem, adds a source and sink with unit source/sink capacities, and performs T shortest-path augmentations. Each internal arc has capacity $T + 1$; this is strictly larger than the total flow, so it never removes an original forward arc from the final residual graph. Parallel arcs from repeated global vertices are retained; they encode different local heights.

There are $2T + V + 2 \leq 6T + 2$ nodes and $10T$ forward arcs. Bellman–Ford uses $O(NE)$ relaxations per augmentation, giving $O(T^3)$ integer relaxations overall, followed by $O(T^2)$ work to extract the potential. If $H = \max(1, |h_{t,i}|)$, the rewards have magnitude at most H , and simple-path distances have magnitude $O(TH)$. The resulting potential can be translated by a constant so that its smallest entry is zero; all entries still have magnitude $O(TH)$. Thus all arithmetic uses $O(\log H + \log(T + 1))$ bits. With ordinary binary addition/comparison, this gives the conservative bit bound

$$O(T^3(\log H + \log(T + 1))),$$

up to vertex-indexing and input/output costs. The T matching rows use $O(T \log(T + 1))$ bits; the potential and normal coordinates use $O(T(\log H + \log(T + 1)))$ bits. No capacity or iteration count depends on the numerical value of a height or a normal coordinate.

A weighted variant. Positive integral weights w_t on tetrahedral discs replace supply and demand 1 by w_t . The same network formulation and integrality proof apply. A general polynomial-time min-cost flow algorithm handles binary weights. The present code intentionally implements only the unweighted case: replacing its T unit augmentations by $\sum_t w_t$ unit augmentations would be pseudopolynomial in binary weights.

3.3 Edge coherence and deletion of zero classes

Call a cooriented normal surface *edge coherent* when all of its intersections with any one oriented global edge have the same sign. An edge with no intersections is allowed. Every surface $S(c_f)$ above is edge coherent.

Lemma 3.2 (Reconstruction from a coherent edge cocycle). *Let S be an edge-coherent cooriented normal surface. Its signed edge intersection numbers form an integral cocycle c_S , and its normal coordinates equal those of $S(c_S)$.*

Proof. The algebraic intersection count around a triangular face is zero: each oriented normal arc contributes cancelling endpoints. Hence the signed edge counts form a cocycle. Edge coherence implies that the unsigned count on edge e is $|c_S(e)|$.

Here is an explicit uniqueness argument for admissible normal coordinates with specified unsigned edge counts. In a tetrahedron, write x_i for its four triangle counts and q_{01}, q_{02}, q_{03} for its three quadrilateral counts, indexed by the pair separated from its complement. The six edge counts give a linear map from these seven coordinates. Its kernel is one-dimensional, generated by

$$(1, 1, 1, 1, -1, -1, -1).$$

One can verify this directly: all four triangles together meet each edge twice, as do all three quadrilaterals together; solving the six homogeneous edge equations gives only this relation. Suppose two admissible coordinate vectors had the same edge counts. Their difference is a multiple λ of this vector. If $\lambda > 0$, the second vector must have all three quadrilateral entries at least λ , violating quadrilateral compatibility; if $\lambda < 0$, the same contradiction applies to the first vector. Thus $\lambda = 0$. Both S and the local-height surface $S(c_S)$ are admissible and have the same unsigned edge counts, proving equality. $\square$

Theorem 3.3 (No zero-class component union at an optimum). *Let f minimize (CS1) for a cocycle c in a compact orientable 3-manifold. No nonempty union of components of $S(c_f)$ represents zero in $H_2(M, \partial M; \mathbb{Z})$. In particular every component is nonseparating within its ambient connected component of M .*

Proof. Suppose a nonempty component union Q represents zero. The complement surface $S' = S(c_f) \setminus Q$ is still cooriented, normal, and edge coherent, because deletion does not create a sign disagreement. The signed edge cocycle of Q represents its Poincaré-dual class, so it is an integral coboundary δg . The remaining surface has cocycle $c_f - \delta g = c + \delta(f - g)$. By the reconstruction lemma, its normal coordinates are precisely those of $S(c + \delta(f - g))$.

Every nonempty normal surface contains at least one disc. Deletion of Q therefore strictly reduces the objective (CS1), contradicting minimality. Finally, a connected cooriented separating properly embedded surface is the relative boundary of a component of its complement, so it represents zero and is excluded by the result just proved. $\square$

Corollary 3.4 (Connected rank-one primitive seed). *Suppose M is connected, $H^1(M; \mathbb{Z}) \cong \mathbb{Z}$, and $[c]$ is a primitive generator. Every minimum-disc cocycle seed $S(c_f)$ is connected and nonseparating.*

Proof. Every component is nonseparating by the theorem. A connected cooriented nonseparating surface represents a primitive relative homology class: join a point on one side to a point on the other through its connected complement and close across the surface. The resulting closed curve has algebraic intersection number one with that surface. In rank one, each component therefore represents either the positive or the negative generator. Opposite signs would give a nonempty two-component union with zero relative class, which the theorem excludes. All signs agree, and their sum is the given primitive generator, so exactly one component is present. $\square$

Checking the hypotheses. The arbitrary-height API certifies only the optimization problem. The face-pairing adapter checks reciprocal gluings, integer height transitions, and normal matching; it does not certify manifold links. The archive02 protocol adapter uses its separately validated finite triangulation object. The connectedness theorem requires both rank one and integral primitivity. In a spanning-tree gauge, integral cocycles form the integral kernel of a matrix and hence a saturated sublattice. When that kernel has rational dimension one, a nonzero kernel vector whose coordinate gcd is one generates its integral lattice. This provides a simple primitivity check for that specific rank-one construction. An arbitrary rational cohomology basis in higher rank is not thereby a saturated integral basis.

The initial spanning-tree gauge already has a related property: a zero-class component union would give an exact edge cocycle vanishing on every tree edge. Its vertex potential is then constant, so its cocycle is zero; edge coherence forces the union to be empty. Thus the connected rank-one seed is not a new existence theorem. The optimization result shows that the guarantee survives leaving this special gauge, although arbitrary changes of vertex potential can introduce separating components.

3.4 What this supplies to a hierarchy implementation

Theorem 6.2 of Lackenby’s 2026 preprint [7] constructs nonseparating normal surfaces from integral cocycles. The present routine supplies an exact minimum-disc representative within the vertex-potential family of a selected class, with an independent lower-bound certificate. The general principle that suitable topological optimization problems become integral linear programs is classical; for example, Dey, Hirani, and Krishnamoorthy [3] treat optimal homologous cycles. The specific local-span objective, binary implementation, and no-zero-component-union consequence above are the contribution being documented here, without a claim of literature-wide priority.

There is a small normalization clarification in the proof of Lackenby’s Theorem 6.2, on page 27 of the PDF. After reducing a rational kernel to one dimension, the condition that the sum of its coordinates is one need not intersect that line: the line spanned by $(1, -1)$ is a simple example. Set one known nonzero coordinate to one instead, then clear denominators. This is an implementation-safe repair of that linear-algebra step; it does not challenge the stated theorem. The normalizing row then has norm one, so the existing Hadamard determinant estimate still applies.

Limits and further questions. The optimization does not minimize pattern-complexity or prove incompressibility. A lower disc count is not a demonstrated reduction of the hierarchy’s number of resets. It supplies neither compressed cutting with all attaching data nor a bound on hierarchy depth. In a one-vertex triangulation, every local height receives the same global shift and the objective is constant; the code returns a certificate of this limitation. Truncation or subdivision can create useful vertex freedom, but increasing the triangulation also changes the

objective and costs work. No claim of subdivision-invariant improvement or of completeness over all normal surfaces is justified.

The genuine finite-exterior experiments therefore use the ten-vertex, 56-tetrahedron triangulations produced by truncating the two-tetrahedron ideal trefoil and figure-eight exteriors. Simplifying those triangulations first yields one-vertex triangulations and removes this particular freedom. For the spanning-tree cocycles tested, the disc counts decrease from 109 to 54 and from 60 to 38. Independent Regina checks give connected orientable surfaces of Euler characteristic -1 with one boundary component. These are measurements of a normal-seed primitive on two examples, not end-to-end recognition speedups or evidence of a general asymptotic bound.

Useful next questions are whether the potential can be optimized jointly with a small family of cohomology classes, whether subdivision choices can be guided by a certified benefit/cost estimate, and whether disc optimization can be combined with a separate certified pattern-complexity descent. A weighted objective can favor expensive local handle types, but its network algorithm must remain polynomial in binary capacities.

3.5 A compressed Euler-characteristic check

The normal coordinates also permit a useful positive-certificate test without expanding the discs. Let $F = D(f)$, let A be the sum of the numbers of sides of all normal discs, and let A_∂ be the total number of normal arcs lying in unpaired boundary faces. If $q_{t,k}$ and $x_{t,i}$ are the quadrilateral and triangle coordinates, then

$$A = 3 \sum_{t,i} x_{t,i} + 4 \sum_{t,k} q_{t,k}.$$

Each internal normal arc is counted twice before gluing, and each boundary arc once. Each point on a global triangulation edge is a vertex of the surface cell structure. Thus

$$\chi(S(c_f)) = \sum_{e \text{ global}} |c_f(e)| - \frac{A + A_\partial}{2} + F. \quad (7)$$

All terms use binary coordinates and a linear number of incidence records. The formula assumes the verified manifold triangulation and normal matching already required above; it is not a way to infer those hypotheses from an arbitrary list of heights.

Proposition 3.5 (A sufficient positive test). *Suppose M has independently been identified as the exterior of the input knot, and the primitive rank-one hypotheses for the optimized seed hold. If the value in (7) is one, the knot is trivial.*

Proof. The seed is a connected orientable properly embedded surface. The classification formula $\chi = 2 - 2g - b$ shows that $\chi = 1$ forces $g = 0$ and $b = 1$, so it is a disk. Its nonzero primitive relative homology class implies that its boundary is an essential longitude on the boundary torus of the knot exterior. Joining that longitude to the core knot by the corresponding annulus in its tubular neighborhood produces an embedded spanning disk. $\square$

This is a proposed certificate interface, not a newly integrated recognition stage. A value at most zero is inconclusive. Even for the unknot, minimizing disc count within one cocycle's vertex-potential family has not been shown to find a disk. In particular, the optimizer's globality is globality within the stated family, not among all normal representatives of the homology class. The package's product-solid-torus examples illustrate a successful case, while the two knotted exteriors give $\chi = -1$.

3.6 Transporting an optimum through known gauge changes

The dual certificate has another direct performance consequence on a fixed triangulation. Suppose new local heights are known to have the form

$$h'_{t,i} = h_{t,i} + g(v_{t,i}) + a_t,$$

where g is a global integral potential and a_t is an arbitrary integral offset for tetrahedron t . These changes describe the same cocycle class.

Proposition 3.6 (Certificate transport). *If f and a matching certify the optimum for h , then $f' = f - g$ and the same matching certify the same optimum for h' . The normal coordinates are unchanged. Transport and verification take a linear number of binary arithmetic operations after indexing.*

Proof. The adjusted heights become $h'_{t,i} + f'(v_{t,i}) = h_{t,i} + f(v_{t,i}) + a_t$. Every local span and gap is unchanged. On a matched row (t, i, s, j) , the common global vertex cancels the added g values, leaving an added reward $a_s - a_t$. Sources and targets are each permutations of the tetrahedra, so these offsets sum to zero. The old primal/dual equality therefore remains valid. The reported initial disc count must of course be recomputed for the new input before serializing a certificate. $\square$

This transport formula is a proved opportunity for a future cache interface; the patch does not expose a new transport API. It also sharpens the interpretation of the large-coboundary experiment: if the generating potential and an old optimum are supplied, one can transport that optimum without solving a new flow. The experiment deliberately asks the general optimizer to recover an optimum from the changed heights alone. Cutting or retriangulating changes the incidence data and is not covered by this formula.

4 Strict-majority witnesses in compressed relator search

The group route represents relators by shared straight-line grammars. Their expanded lengths can be enormous even when a certificate and each local word operation remain small. The baseline already has exact compressed equality, slicing, inverse, longest-common-prefix (LCP), and longest-common-substring (LCS) operations. The present improvement changes what the search asks the LCS primitive to prove. It leaves certificate replay unchanged.

4.1 The amount of overlap needed for shortening

After choosing cyclic rotations and possibly inverting the donor, write

$$D = UA, \quad R = UB,$$

where D is a donor relator, R is a target relator, and U is the common prefix. Replacing R by $A^{-1}B$ is an elementary relator operation: $D^{-1}R = A^{-1}B$ before free reduction. Let $m = |D|$, $r = |R|$, and $\ell = |U|$. Before any extra cancellation, the replacement has length $m + r - 2\ell$, so the raw reduction in total relator length is

$$\text{gain} = 2\ell - m. \tag{8}$$

Strict raw shortening therefore requires only

$$\ell \geq \tau(m) := \lfloor m/2 \rfloor + 1.$$

Finding the globally longest common cyclic factor is stronger than necessary to find one valid shortening move. The optional policy asks first for a witness of length $\tau(m)$, then improves that witness's selected alignment.

The word “shortening” in the next theorem refers to this strict raw overlap criterion. Additional cancellation can help a particular relator move, but the theorem does not assert completeness for every possible group-theoretic reduction involving such cancellation or other operations.

Theorem 4.1 (Local existence is preserved). *Replace the exact cyclic LCS query's old shared-count cap by $\tau(m) = \lfloor m/2 \rfloor + 1$. Search both signs of the donor, and use the shorter member of each unordered relator pair as donor. With unlimited resources, the modified query finds a strict raw-shortening overlap if and only if one exists among the examined relators and their cyclic rotations.*

Proof. The exact capped LCS primitive returns the smaller of the true optimum and its supplied cap, together with checked matching positions. Thus returning the threshold is equivalent to existence of an overlap above half the donor length. Doubling each cyclic spelling contains every one-turn cyclic factor; starts are reduced modulo the original lengths. Restricting overlaps to at most m ensures that the resulting witness is a genuine one-turn factor.

Both donor signs cover relative orientation. If an overlap of length ℓ shortens with the longer member as donor, then $2\ell - |D_{\text{short}}| \geq 2\ell - |D_{\text{long}}| > 0$; the shorter-donor choice therefore preserves existence. The existing absolute letter-count bound is an upper bound on any one-turn overlap, so its rejection cannot discard a strict-majority witness. These facts cover every permitted pair, rotation, and relative sign. Conversely, every returned witness is checked and satisfies (8) with positive gain. $\square$

This is a local existence theorem. It does not say that a greedy sequence of these moves decides every unknot presentation, nor that it preserves the later trajectory chosen by the previous longest-overlap policy.

4.2 Two exact extensions make the selected alignment maximal

Suppose a length- ℓ witness starts at i in the target and j in the donor, with $m \leq r$. Normalize both offsets to one turn. In doubled roots, compare the unmatched intervals beginning at $i + \ell$ and $j + \ell$, each truncated to at most the remaining $m - \ell$ letters. One LCP query extends to the right by a letters. Let $q = \ell + a$.

If $q = m$, the donor-length cap is reached. Otherwise compare the preceding $m - q$ letters backwards by taking the LCP of their inverses. If this adds b letters, return the factor of length $q + b$ at starts

$$i - b \pmod{r}, \quad j - b \pmod{m}.$$

Taking inverses reverses order and reverses signs in both words, preserving equality of the compared preceding factors.

Lemma 4.2 (Exact alignment extension). *The procedure contains the original witness, uses at most two exact LCP queries, and returns a common cyclic factor maximal at its selected relative alignment subject to the one-donor-length cap. If its length is less than m , the two boundary letters disagree on each side.*

Proof. The forward LCP stops at its first disagreement or its length cap. If it stops at a disagreement, extending backwards does not change that right boundary. The backward LCP stops at the first left disagreement or at the remaining donor-length allowance. Hence an uncapped side cannot be extended further. The original matched interval is retained throughout and every

new letter is certified by an exact LCP equality. The total length never exceeds m , so cyclic starts remain valid. $\square$

The implementation uses a constant number of slices and inversions, and all operations share the same arena work allowance and cancellation callback. Slices add only a number of grammar rules controlled by grammar height; inversion traverses a DAG. Since the existing exact LCS and LCP primitives have polynomial bit bounds in grammar size and length bit size, the new procedure retains that local bound. No expansion of an exponentially long word or list of all its positions is introduced.

4.3 A family with a proved exact optimum

Let

$$P = (ab)^N, \quad T = \text{aabbccacbab}, \quad N \geq 16,$$

and consider the positive cyclic relators

$$D = P cTa, \quad R = P aTb. \quad (9)$$

Each has length $2N + 14$. The fixed word T contains all nine directed bigrams on $\{a, b, c\}$, so the previously added alphabet and transition filters do not distinguish the two words. Their letter-count vectors differ by one c versus one b . Since they have equal length, neither whole donor can occur in a rotation of the other. The inexpensive whole-donor count rejection therefore bounds the work of the stage that precedes partial overlap search.

Proposition 4.3 (Exact cyclic LCS for the stress family). *The maximum one-turn cyclic overlap of the two words in (9), allowing both relative signs, has length $2N$. Both the historical and the optional policy can attain this overlap. The resulting replacement has length 28, with gain $2N - 14$.*

Proof. The common prefix P proves a lower bound $2N$. An inverse of a positive word has only negative letters, so the opposite-sign case has no nonempty overlap here.

In D , the unique alternating a, b run of length at least four is P . It is preceded by a , giving aa at the left boundary, and followed by c . In R , the unique long alternating run is bPa , of length $2N + 2$; it is preceded by c and followed by a , giving aa at its right boundary. All other alternating runs have length at most three.

If a common cyclic factor had length at least $2N + 1$, it would contain at least $2N - 13$ letters of D 's principal run, split into at most two intervals. For $N \geq 16$, one interval has length at least ten. It must align with the principal alternating run of R . Such a common factor cannot cross the left end of P in D : either it first encounters R 's preceding c while D is still alternating, or D 's preceding a disagrees with the b forced by R 's alternation. It cannot cross the right end either: D 's next c disagrees with the continuing alternating run of R , or the aa departure from R 's run first disagrees with D . The factor is therefore contained in D 's P and has length at most $2N$, a contradiction.

At that optimum the replacement is $(cTa)^{-1}(aTb)$. Its two parts each have length 14 and their boundary letters cause no free or cyclic cancellation. Its length is 28, giving the stated gain. $\square$

The threshold is only $N + 8$. On the prefix presentation, the threshold witness is obtained immediately and the forward extension recovers all of P ; the opposite cyclic boundary disagrees. To avoid relying only on a common-prefix fast path, the tests also rotate the spellings to $cTaP$ and $aTbP$. The shared run is then an interior/suffix witness found through the compressed aligned-cut route.

For $N = 2^h$, the constructed grammars have $O(h)$ rules. The supplied positive instances exercise no full occurrence table: the prefix case requires no cut-pair visit, and the rotated case uses one aligned-cut extension. The needed counts, shared subgrammar comparisons, slices, and inverses have polynomial cost in h . Tests prohibit both word expansion and occurrence-table construction on these inputs through $h = 500$. This establishes a meaningful compressed-family improvement; it is still a statement about the specified local search and application operation.

A negative control that matters. The tempting family PcT, PaT is not a corresponding hard cyclic example. The common tail T wraps before P , creating an overlap of length $2N + 12$ that the older method already finds cheaply. The benchmark retains this family as an overhead control. Treating it as a hard linear-string instance after cyclic doubling would have overstated the improvement.

4.4 The policy has no uniform greedy-gain guarantee

Exact extension does not make an arbitrary threshold witness globally best. Let the donor consist of $2m$ distinct letters. Put its first $m + 1$ letters at the start of a target, followed by fresh separator letters, then its first $2m - 1$ letters, then another fresh separator. The first witness is locally maximal with overlap $m + 1$ and gain 2. A different alignment has overlap $2m - 1$ and gain $2m - 2$.

Proposition 4.4 (Unbounded loss of greedy gain). *The ratio of the first maximal threshold witness’s gain to the best gain can be $1/(m - 1)$, tending to zero. There is no positive universal gain approximation factor for the present first-witness policy.*

Proof. The fresh separators prevent either displayed interval from extending. Every donor letter is distinct, so a common cyclic interval cannot pass a separator or combine the two copies. The longer displayed copy realizes the optimum, and (8) gives the two gains and their ratio. $\square$

The concrete test uses donor `abcdefgh` and target `abcdeXYZabcdefgW`: the old overlap is seven, while the optional overlap is five. Duplicating every letter removes singleton occurrences without removing the discrepancy; the independently computed weighted Whitehead cut on that doubled example has value zero. The concern therefore is not confined to presentations immediately eliminated by a singleton rule.

The historical policy remains the default. A future scheduler could keep cheap witnesses while spending a bounded additional budget on improving their gain, but any approximation or amortized descent theorem would require an argument beyond the local existence theorem above.

4.5 Certificates, real residuals, and scope

The helper emits the existing relator-move format. Literal and compressed certificate verifiers independently check the cyclic positions, orientation, overlap, and replacement. They do not invoke the producer to justify its own move. The new search policy does not alter the trust boundary.

To test a genuine knot-derived state, the audit replays the first 132 moves of the archived Gordian certificate. Its ten remaining nonempty relators have lengths

$$378, 8140, 4, 370, 1704, 370, 2026, 128, 888, 1022.$$

The optional query selects target slot 124 and donor slot 73, rotations 81 and 3, positive donor sign, and overlap four. The continuation gives the same 142-move original-diagram certificate as the previous policy. Both independent replayers accept when producer/helper calls are disabled,

and both reject a forged overlap. This tests a reachable certificate path; it does not demonstrate a speedup of the whole recognition query.

A separate attempt to force the compressed-overlap route from the start of Gordian exhausted twenty million work units. The ordinary corpus routes through the existing explicit bridge at these sizes. Moreover, the no-majority pair $(ab)^N, (aabb)^N$ can exhaust its allowance in the preceding whole-donor probe before the new helper is called. These observations identify next bottlenecks and prevent an isolated successful kernel from being mistaken for an improved complete search algorithm.

5 Correctness checks and controlled measurements

The experiments serve three purposes: detect implementation errors against independent calculations, identify the source of an improvement, and find conditions under which an improvement fails to help. They do not fit an asymptotic exponent to a small sample and do not infer a worst-case theorem from fast synthetic instances.

5.1 Environment, timing scope, and censoring

The principal runs use Python 3.12.14 on Linux x86-64 with glibc 2.39. Independent geometric and polynomial checks use Regina distribution 7.4.1, whose engine version string is 7.4 [1]. The package records full runtime strings and source SHA-256 hashes. This is a shared execution environment, without a claim of dedicated CPU isolation. Small differences in sub-millisecond medians should be interpreted accordingly.

The tensor comparison uses seven measured rounds and one excluded warmup per case and arm. Within each round it shuffles six arms using seed 2645: faithful Potts, a second identical faithful control, certified Laurent tensor, certified global-shift integer tensor, certified valuation-normalized integer tensor, and an ordinary-order integer tensor. Every query starts with a fresh validated diagram, computes the complete polynomial, and includes order preparation. The allowance is 4,096 states, 200,000 transitions, and two seconds per query. The ordinary-order arm is an ordering ablation and has no universal width promise.

The overlap comparison uses five measured shuffled rounds, one excluded warmup, and seed 3110. It runs the archived helper from the pinned revision, an A/A repeat, and the optional helper in a common harness. Local timings include grammar construction, the preceding whole-donor probe, the cyclic query, and checked application. The local allowance is two million charged work units and 100,000 nodes. Whole recognition timings include fresh input validation and independent certificate replay. The exact benchmark options and every raw sample are retained.

An exhausted allowance is recorded as `LIMIT`, never as a completed negative query. It has no completed-query speedup ratio. A shorter time to hit a limit can be worse performance, because it has produced no answer. Likewise, an earlier but smaller greedy move is not interchangeable with a maximal-gain move merely because both shorten a relator.

5.2 Tensor correctness and full-polynomial timings

The independent tensor audit performs 420 full-polynomial comparisons for each of the three arithmetic modes, for 1,260 comparisons in total. It also enumerates 23,676 smoothing states and follows 71,860 oriented smoothing circles independently; every absolute turn is four. The cases exercise mirrors, row permutations, signed arc relabelings, supplied orders, different outer faces, and replay of separator certificates. These totals count audit operations, not 1,260 distinct knots. A separate program imports each diagram into Regina with independently checked component

count and writhe. Regina’s polynomial variable is $\sqrt{t}$, so its even exponents are divided by two, without a mirror inversion. It checks 35 base inputs and their mirrors in all three tensor modes, giving 210 further complete coefficient comparisons. Named examples, deterministic braid closures, weaving diagrams through 24 crossings, and the two residual-twist witnesses of Section 1.4 are included. The independent conversion uses a closed binomial formula for the latter formal-coordinate check, rather than the baseline’s recurrence.

The final six-arm timing run contains 630 measured queries: 588 complete and 42 capped, plus 90 excluded warmups. Every completed polynomial agrees across the applicable arms. Table 2 reports the full 15-case corpus; the code and raw data specify every input diagram.

Case	n	Faithful (ms)	A/A (ms)	Laurent (ms)	Global integer (ms)	Normalized integer (ms)	Ordinary integer (ms)
Trefoil	3	0.107	0.120	0.273	0.260	0.278	0.179
Conway	11	0.425	0.403	1.512	1.232	1.347	0.758
Kinoshita–Terasaka	11	0.396	0.417	1.508	1.234	1.322	0.776
Hard unknot	8	0.229	0.227	0.602	0.586	0.593	0.447
Five-strand stress	36	2.013	2.316	9.664	6.677	6.699	4.498
Weaving $q = 5$	10	0.419	0.411	1.412	1.234	1.326	0.818
Weaving $q = 13$	26	1.209	1.177	6.393	4.478	4.031	2.608
Weaving $q = 17$	34	1.967	1.820	9.839	5.483	5.428	3.313
Tree, depth 6	126	8.066	7.824	25.659	37.402	24.664	LIMIT
Tree, depth 7	254	67.007	64.860	72.620	242.443	68.276	LIMIT
Grid 4×4	16	0.473	0.461	1.704	1.724	1.694	1.039
Grid 6×6	36	1.434	1.411	5.441	4.977	5.172	3.183
Grid 8×8	64	4.554	4.478	13.699	13.050	12.090	8.255
Grid 10×10	100	30.232	33.614	43.047	43.681	35.147	25.901
Shuffled 8×8	64	20.268	20.260	LIMIT	LIMIT	LIMIT	LIMIT

Table 2: Complete full-Jones queries: median milliseconds over 7 randomized paired rounds after 1 warmup round. A/A is an independently timed repeat of the faithful arm. Order construction and verification are included; the ordinary-integer arm omits the universal width certificate. LIMIT denotes a capped arm, not a completed-query runtime; no ratio is assigned to it. Per-query limits were 2 seconds, 4,096 states, and 200,000 transitions. These invariant-query measurements do not measure complete recognition.

The faithful implementation remains faster on most of these inputs. This does not conflict with the improved universal bound: equality partitions can collapse to very small tables on favorable actual diagrams. For example, the recorded Conway calculation has five faithful states versus nineteen binary tensor states; the favorable shading of the 254-crossing tree medial has one faithful state versus 272 tensor states. A smaller asymptotic worst case does not force a smaller table on every instance.

Within the tensor implementations, valuation-normalized integers improve several cases. The median *paired* Laurent/normalized ratios are 1.885 on the 34-crossing weaving case, 1.432 on the 36-crossing stress braid, 1.151 on the 100-crossing grid, and 1.072 on the 254-crossing tree medial. The trefoil ratio is 0.994, providing no improvement. Paired ratios need not equal ratios of the separately reported median times.

The global-shift arithmetic ablation reveals a specific avoidable cost. On the 254-crossing tree medial it retains 130,177-bit scalars and takes 242.443 ms, whereas valuation normalization retains 1,536-bit mantissas and takes 68.276 ms. Both compute the same exact polynomial, with the same binary frontier method. Table 3 and Figure 1 show the corresponding precision counts.

Case	n	Faithful scalar bits	Laurent coefficient bits	Global integer bits	Normalized mantissa bits
Tree, depth 6	126	63,757	2	32,321	768
Tree, depth 7	254	258,573	2	130,177	1,536
Grid 10×10	100	10,855	2	10,921	832

Table 3: Maximum bit length of one stored scalar, Laurent coefficient, or normalized mantissa in completed queries from the final paired run. The values agree across all repeated samples. They are not total table-memory measurements: the normalized mode also stores a valuation, and every mode stores frontier keys and other metadata. The two integer tensor modes use the same exact polynomial model and certified order.

Stored integer precision

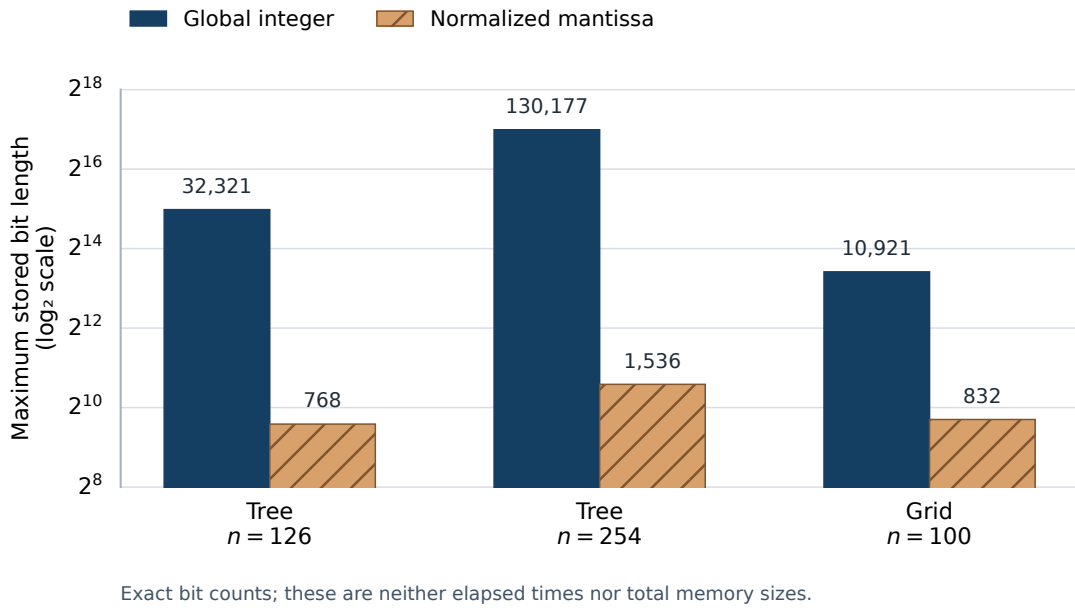


Figure 1: Retained scalar precision in the same exact tensor contraction, with and without valuation normalization. The axis is logarithmic and the labels give exact bit counts. This measures monomial-padding removal, not an asymptotic running-time fit or a change in frontier width.

The ordering controls also matter. The certified tensor orders complete the two large tree cases where ordinary-order integer tensors hit the state cap. Conversely, every tensor mode hits the cap on the shuffled 64-crossing grid, while faithful Potts completes. Thus the new backend adds a stronger exact general bound and an alternative performance profile; it does not dominate the existing backend on the supplied finite corpus.

5.3 Cocycle seeds: exact objectives and genuine manifolds

The cocycle data contain seven explicit product-solid-torus cases, a one-vertex numerical limitation example, and two genuine finite knot exteriors. The product construction triangulates $D^2 \times S^1$ by staircase tetrahedra. It starts with the primitive circle cocycle and adds a large global vertex coboundary. Hence the huge initial weight represents the same fixed primitive class; this deliberately isolates dependence on binary height size.

T	Height bits	bits(D_0)	D_*	Optimizer (ms)	Checker (ms)
9	1	2	3	0.812	0.125
9	20	23	3	0.657	0.111
9	260	263	3	1.035	0.134
9	4100	4103	3	1.163	0.128
24	20	25	3	3.599	0.250
48	20	26	3	15.221	0.498
96	20	27	3	46.796	0.954

Table 4: All seven product-solid-torus cases. T is the number of tetrahedra; D_0 and D_* are the initial and optimal numbers of normal discs. Height bits are measured from the supplied local heights, rather than inferred from a case-name parameter. Times are single recorded calls, not paired benchmark medians. Optimization includes its internal certificate check; the checker column measures an additional certificate and face-matching check. These cases verify binary-height handling and the fixed-family optimum; they do not establish an end-to-end recognition speedup.

The largest-height nine-tetrahedron case starts with a disc count having 4,103 bits and returns a three-disc meridian seed. Its flow still takes exactly nine augmentations. It therefore tests the absence of loops proportional to numerical weight. The timings in this table are individual recorded observations, not paired medians or a random-knot study. In the one-vertex numerical case, the initial and optimized counts remain equal; the optimizer correctly certifies that a common vertex shift changes no span. No manifold is asserted for that arbitrary-height one-row example.

For a geometric test independent of the artificial large coboundaries, the audit truncates two-tetrahedron ideal trefoil and figure-eight exteriors to finite triangulations. Each has 56 tetrahedra and ten vertices. The existing archive02 exact spanning-tree routine supplies a cocycle; rank one and a coordinate gcd of one verify integral primitivity. The optimizer does not choose a different homology class.

Knot exterior	T	$ V $	D_0	D_*	Optimizer (ms)	χ	b
Trefoil	56	10	109	54	19.655	-1	1
Figure-eight	56	10	60	38	20.832	-1	1

Table 5: Two genuine finite knot exteriors obtained by truncating Regina’s ideal examples. A primitive tree-gauge cocycle is optimized within its vertex-potential family. Regina independently verifies that each output is connected and orientable; χ and b denote its Euler characteristic and boundary-component count. The single-call times include the triangulation adapter’s seed construction and consistency checks, but exclude triangulation construction, cohomology preparation, and the independent Regina check. Disc-count optimality in this family does not assert genus minimality or incompressibility.

The disc counts fall from 109 to 54 and from 60 to 38. Regina independently checks valid orientable finite manifolds and connected orientable surfaces of Euler characteristic -1 with one boundary component. Exact gluings, isomorphism signatures, cocycles, normal coordinates, and primal/dual certificates are included in the JSON. Simplifying these exteriors first produces one-vertex triangulations and removes this optimization freedom; the choice to optimize the finite truncations is explicit.

Fifteen focused tests also compare small instances against exhaustive primal searches and dual permutation oracles, exercise repeated vertices and cancellation, reject corrupt certificates, handle 4,096-bit heights, and check archive02 interoperability. The independently verified equality of primal and dual objectives is the evidence of optimality. A decreased disc count by itself is not

evidence of faster hierarchy construction.

5.4 Overlap measurements, controls, and unchanged corpus paths

Family, $N = 2^h$	Old ms	A/A ms	Witness ms	Witness work	Nodes
Prefix, $h = 8$	174.971	161.297	0.551	708	64
Prefix, $h = 32$	LIMIT	LIMIT	0.731	1,308	112
Prefix, $h = 500$	LIMIT	LIMIT	5.220	13,008	1,048
Interior, $h = 8$	201.398	159.737	0.503	962	76
Interior, $h = 32$	LIMIT	LIMIT	1.239	1,898	148
Interior, $h = 500$	LIMIT	LIMIT	8.100	20,150	1,552
Shared tail, $h = 500$	7.012	5.772	6.916	12,939	1,042
Gain tradeoff	0.240	0.231	0.190	488	43

Table 6: Five-round medians for complete local search and application. The strict-majority family has the same final overlap and gain in each successful arm. The gain-tradeoff row intentionally has different move quality. LIMIT means exhaustion of two million charged work units.

On the $h = 8$ prefix and interior cases, old charged work is respectively 245,591 and 319,306 units. The optional counts are 708 and 962. For larger h , the old arms exceed the allowance, so no numerical speedup factor is reported. The retained shared-tail control shows no convincing gain, as predicted by the cyclic-factor analysis. In the no-majority $h = 500$ control, all arms exhaust in whole-donor search with zero witness-helper calls.

All 225 measured whole-knot queries retain identical certificate digests, and none reaches the changed compressed LCS helper. The sums of medians for fourteen smaller inputs are 140.334, 140.396, and 138.959 ms for old, A/A, and optional arms. Gordian medians are 1.5253, 1.5104, and 1.4790 seconds. These numbers support no knot-corpus speedup claim for the new helper.

The reachable Gordian residual produces the same first move, 142-move certificate, and 4,280 allocated nodes in each arm. Charged work changes only from 155,142 to 155,168. Its five-round medians, 69.55, 63.42, and 87.21 ms, fluctuate more than the small work difference would suggest and show no stable improvement. The complete trace is supplied because it validates the real certificate boundary, not because its timing favors the new method.

5.5 Maintained test inventory and the worker correction

Final discovery contains 721 tests in 83 maintained modules. The recorded validation covers this inventory through ordinary `unittest` module runs, with Regina distribution 7.4.1 enabled and no skipped tests. Before the worker repair, all 720 then-existing tests ran: 719 passed and the unchanged large-input worker regression raised a timeout. That error reproduced in isolation. It was a compatibility defect in partial-input communication on the inspected Python 3.12.14 runtime, rather than evidence of a mathematical failure in any of the three primitives.

Only `fastunknot/normal_surface.py` and its test module changed after that comprehensive batch. The other 82 modules supply 712 passing tests; the affected worker module subsequently passes all nine tests, including one added cleanup regression. Thus the final coverage is $712+9 = 721$ current tests. A further six-test public-integration rerun also passes; those six belong to the retained 712 and are not counted twice. The affected module is the only test module using the worker transport; the cocycle adapter tests import Regina directly. No existing test’s deadline or expectation was relaxed. The complete accounting, final source hashes and raw logs are in `validation/final-validation.json` and the adjacent validation files.

An earlier single-process discovery attempt ended with exit code 1 after 14 passing lines, without

a unittest summary or traceback. Its cause was not established; the next test passed in isolation. This attempt is preserved and classified as incomplete, separately from the comprehensive module runs. We do not report a successful final single-process suite run.

The checks distinguish discovery from verification. Seed-certificate verification runs without the flow solver. Group-certificate replay runs without the producer. Tensor output is compared to both an independent state-sum oracle and an external exact implementation. Resource tests ensure that incomplete operations cannot publish a partial invariant or a false decision. Those boundaries are more consequential than simply increasing the number of test cases.

5.6 Interpretation of the evidence

The empirical conclusions are deliberately specific. The binary tensor backend improves a proved repository asymptotic bound. Valuation normalization removes measured arithmetic overhead within that backend. Cocycle flow reduces a certified geometric objective on two actual knot exteriors and handles huge binary heights without expansion. Threshold-first overlap substantially accelerates a proved compressed stress family. None of these observations demonstrates a quasi-polynomial growth law for complete recognition, and the supplied negative controls identify concrete reasons why such an inference would be invalid.

6 Integration, certificate boundaries, and resource behavior

The source patch keeps each optional experiment available through a supported interface. The new production modules have no mandatory third-party Python dependency. Regina is used for optional geometric checks and the existing external fallback, not for either the tensor implementation or the cocycle flow solver.

6.1 Using the exact tensor backend

From the `fast` directory, the standalone command computes a full polynomial and reports its one-sided recognition consequence:

```
python -B -m fastunknot jones examples/trefoil.json --backend tensor
python -B -m fastunknot jones examples/hard_unknot_8.json --backend tensor
```

The first input has a nonidentity polynomial; the second has identity and the standalone Jones command is inconclusive about knot type. To use the tensor stage within the full recognizer:

```
python -B -m fastunknot recognize examples/hard_unknot_8.json \
    --jones-backend tensor
```

Other earlier stages may already decide this small input. Tests explicitly disable those stages when checking that tensor identity continues to the Khovanov fallback, and that a separately exhausted fallback returns UNKNOWN rather than misusing Jones identity.

The low-level interface is:

```
from fastunknot import Diagram
from fastunknot.tensor_jones import tensor_jones

diagram = Diagram.from_braid(3, [1, -2] * 5)
result = tensor_jones(
    diagram, certified=True, arithmetic="integer",
    max_states=4096, max_transitions=200000,
)
```

The result contains the exact full polynomial as signed hexadecimal integer coefficients indexed by integral t -exponent, the chosen order and separator certificate, crossing count, writhe, and measured resource counters. Hex encoding avoids dependence on a runtime's decimal-digit conversion limit for very large exact integers. Callers can select "laurent" for the reference calculation or "integer-global" for the arithmetic ablation. The latter intentionally preserves an inefficient monomial-shift scheme for comparison.

The retained state and transition limits count mathematical work objects. They do not independently cap the bytes occupied by a large integer. Deadline callbacks are checked during preparation, state updates, coefficient loops, and decoding. A running big-integer operation is not preempted at an arbitrary machine instruction, so these are cooperative allowances rather than hard real-time guarantees.

6.2 Optimizing and checking a cocycle seed

The numeric API accepts four global vertex IDs and four integral heights per tetrahedron:

```
from fastunknot.cocycle_seed import (
    minimize_disc_seed, check_seed_certificate,
)

certificate = minimize_disc_seed(vertices, heights)
checked = check_seed_certificate(vertices, heights, certificate)
assert checked["primal"] == checked["dual"]
```

It rejects malformed rows and nonintegral values. The checker recomputes the potential-adjusted local gaps, disc counts, shared-vertex matching, source and target permutations, and objective equality. It does not call the optimizer or trust its reported statistics.

There are two geometric adapters. The face-pairing API validates reciprocal gluings, local integer height transitions, and normal matching. The triangulation-protocol API accepts a separately validated finite triangulation and integral cocycle, including the archive02 object used in the tests. The optimization theorem needs only numeric data; the topological theorems additionally need a genuine compact orientable manifold, coorientation, and the stated integral class conditions. No adapter silently infers manifold validity from an optimal flow.

A normal-coordinate vector together with a primal/dual matching certifies the restricted minimum-disc problem. It does not certify that the surface is incompressible or that a subsequent hierarchy is essential. Those are separate certificate types. The proposed Euler-characteristic positive test in Section 3 would also require independently bound knot-exterior provenance and peripheral data before it could decide an input diagram.

6.3 Enabling the overlap witness policy

The new CLI flag enables its required compressed search and relator moves:

```
python -B -m fastunknot recognize examples/hard_unknot_8.json \
    --group-overlap-witness --group-seconds 2
```

Python callers select the related flags explicitly:

```

result = recognize(
    diagram,
    use_group=True,
    group_compressed_search=True,
    group_relators=True,
    group_overlap_witness=True,
)

```

The standalone compressed producer accepts `relator_moves=True`, `overlap_witness=True`; the local helper accepts `witness_first=True`. Non-Boolean values and inconsistent option combinations are rejected. The adaptive explicit-to-compressed policy is a different interface and is not silently changed by this option.

Search still has the same resource semantics: a verified successful group certificate proves the existing conclusion, while an unsuccessful or capped search remains inconclusive. The first-witness policy may change a future search trajectory, which is why it is not made the default. Neither independent verifier needs to know which search policy discovered a move.

6.4 Repairing large requests to the external normal-surface worker

The maintained test inventory exposed a separate, reproducible compatibility bug in the existing worker transport. The parent initially called `communicate(payload, timeout=...)` and retried with no input after a short timeout. In the tested Python 3.12.14 runtime, the communication code retained its internal input buffer but registered writable stdin only when the current call supplied a truthy input argument. A delayed reader with a 500-kilobyte request could therefore receive only the first pipeful, while the parent waited until its three-second test allowance expired.

The repair gives one communication thread sole ownership of `communicate(payload)`. The caller waits on a completion event in bounded intervals, checking the existing local deadline and global cancellation callback on the calling thread. Completion publishes either the returned streams or the communication error. A worker pipe failure remains inconclusive, while a caller's cancellation exception propagates.

Cleanup covers every operation after successful child creation, including event construction, thread construction and startup. It kills a live child, reaps it, joins a successfully started communicator, and then closes the pipes. Independent review identified and closed a constructor-scope gap before the final tests. The scope is the controlled direct-child Regina worker; it is not a general supervisor for arbitrary grandchildren retaining inherited pipes.

The existing delayed-reader regression now transmits and recovers all 500,000 input characters within its unchanged three-second allowance. Existing cancellation and real-engine checks also pass. One additional test injects communication and thread-setup failures, verifies child reaping and closed pipes, checks that the payload is submitted once, and confirms caller-thread polling. The final nine-test worker module passes in 2.732 seconds, followed by six public-integration tests in 0.010 seconds. The preserved diagnosis and logs are separate from the performance benchmarks; this transport repair contributes no claimed mathematical speedup.

This is an engineering correctness and performance repair, not a new topological theorem. The external engine remains explicitly trusted for its own topology operations. Digest validation, response schema checks, decision semantics, and the complete built-in fallback remain in force.

6.5 What an independent verifier does and does not establish

Evidence	Independent check	Conclusion
Separator hierarchy	Graph partition, balance, order, cut width	The stated width bound for that input and order.
Full tensor polynomial	State-sum and Regina comparison in the audit	Implementation agreement on tested inputs; the general exactness theorem comes from Section 2.
Seed potential and matching	Recomputed local spans and equal dual bound	Global optimality within the vertex-potential family.
Relator trace	Literal or compressed move-by-move replay	The existing group certificate conclusion for the original diagram.
External normal decision	Input digest, process status, schema, engine label	A completed externally trusted decision, not independent replay of a geometric hierarchy.

Table 7: Trust boundaries retained by the integration. A consistency check is not automatically a certificate of a stronger mathematical claim.

All complete output claims refer to a validated input. An invalid planar encoding, unmatched normal coordinates, an inexact division, or a malformed certificate is a failed operation. A resource exhaustion is a different outcome and remains inconclusive. Preserving this distinction is essential when combining several exact and heuristic discovery stages in one recognizer.

7 What is still needed for quasi-polynomial recognition

The most useful next step is to make the missing global bounds explicit enough to prove, refute, or measure. A vague expectation that every local operation is fast is not an adequate complexity argument. In particular, binary coordinates can encode exponentially many discs, a small grammar can undergo exponentially many valid rewrites, and a small boundary alphabet can carry a large algebraic object.

7.1 A complete accounting identity

Let R be the number of processed search or hierarchy states, let S_j be the complete binary encoding size of state j , and let C_j be the cost of generating, transforming, and checking it. If local operations have a uniform polynomial bound $C_j \leq (S_j + N)^d$, then

$$\text{Time}(N) \leq \text{preparation}(N) + R \left(N + \max_j S_j \right)^d. \quad (10)$$

Thus both R and the maximum represented size must be quasi-polynomial. This elementary inequality is the central obligation that the three new primitives do not yet discharge.

For the hierarchy route, the lexicographic reset accounting in [7, Section 9] gives at most $L(g+1)^L$ passes through a step, where L bounds hierarchy length and g bounds relevant pattern-complexities. Taking logarithms shows the precise requirement:

$$\log L + L \log(g+1) \leq (\log(N+2))^{O(1)} \quad (11)$$

would suffice for the iteration part of a quasi-polynomial bound. A linear bound on L together with a polynomial bound on g gives only an $\exp(O(N \log N))$ reset estimate by this argument. Strict decrease of a large integer potential proves termination, but is not the same as a small number of decreases.

The refined hierarchy controls length through its initial handle complexity, but this does not by itself supply (11). A successful program could either obtain much stronger simultaneous depth and complexity bounds, or replace the lexicographic reset estimate with a better amortized or memoized one. The latter may be more plausible than forcing every decomposition to halve a topological quantity. In either case the actual compressed cutting and attaching operations must satisfy the local cost assumption in (10).

7.2 A conditional composition theorem with explicit hypotheses

The following statement is an accounting tool, not a theorem that the current hierarchy or group search satisfies its hypotheses.

Theorem 7.1 (Conditional quasi-polynomial composition). *Suppose a complete recognition procedure on an N -bit input can be organized as a rooted search tree with the following properties, for fixed constants a, b, c, d, C :*

1. *its depth is at most $C \log^a(N + 2)$;*
2. *each node has at most $(N + 2)^c$ children, including every necessary branch of the complete procedure;*
3. *every node, attaching map, integer, and certificate needed locally has total binary encoding size at most $\exp(C \log^b(N + 2))$;*
4. *constructing a node, deciding its local actions, checking its certificate, and verifying a leaf take time at most the d th power of its encoding size plus N .*

Then the complete procedure has quasi-polynomial bit time and can be executed in quasi-polynomial space by depth-first traversal.

Proof. The number of nodes is bounded by a geometric sum with largest term

$$((N + 2)^c)^{C \log^a(N + 2)} = \exp(O(\log^{a+1}(N + 2))).$$

The per-node cost is $\exp(O(\log^b(N + 2)))$ times a polynomial factor in N . Multiplying gives $\exp(O(\log^{\max(a+1, b, 1)}(N + 2)))$. The active path has polylogarithmic length and each stored state has the stated bound; the local workspace is bounded by the local running time. This gives the space conclusion. Completeness is an explicit hypothesis; it is not obtained from this counting argument. $\square$

Another route replaces depth and branching by a proved quasi-polynomial bound on the number of distinct canonical states and a rule that expands each at most once. That requires a canonical representation preserving all attaching and boundary-pattern data. Equal coarse statistics or isomorphic unlabelled incidence graphs are not automatically equivalent states for a future geometric continuation.

For a width-based route, $w = \log^{O(1)} N$ makes the binary tensor primitive quasi-polynomial immediately. Planarity alone only supplies $w = O(\sqrt{n})$. Moreover, exact Jones identity still requires an independent completion route in this implementation. A width theorem and a complete treatment of identity cases would therefore both be needed before that observation became a recognition theorem.

7.3 Safe portfolio accounting

The measurements suggest a practical improvement that already has a simple complexity proof: run faithful Potts for a polynomially bounded amount of whole-query work, and if it does not

finish, run the certified binary tensor algorithm from the validated diagram. The first arm is often much cheaper on the current corpus. The second supplies the better universal bound.

Proposition 7.2 (A bounded prelude preserves a fallback bound). *If an exact prelude costs at most $p(N)$ bit operations and returns either a checked answer or an inconclusive result, while an exact complete fallback costs at most $F(N)$, sequential composition costs at most $p(N) + F(N)$ and is exact. In particular, a polynomially bounded faithful-Potts prelude preserves the binary tensor full-Jones bound.*

Proof. The prelude either supplies a checked answer within its budget or consumes at most $p(N)$ before the fallback starts. The fallback receives the original validated problem. Its cost and correctness then apply independently of the failed prelude. Adding the bounds proves the claim. $\square$

The budget must include preparation, all attempted shadings or orders, coefficient arithmetic, and normalization, not just one inner transition counter. In this implementation those operations have polynomial bit bounds per represented work object, so a proved polynomial transition and state budget can support such a prelude. The portfolio itself is proposed, not part of the frozen patch.

There is an analogous constraint on an eventual quasi-polynomial recognizer: it cannot always insist on finishing an uncapped $2^{O(\sqrt{n})}$ invariant first and then claim a quasi-polynomial total from a later stage. Such an invariant must have its own quasi-polynomial allowance, be skipped, or be proved cheaper on the relevant branch. Optional heuristics are compatible with a good worst-case theorem when their discarded work is explicitly bounded.

7.4 Priority I: complete compressed geometry

Research question 7.3 (Exact cutting with complete attaching data). Can a cut along a binary normal surface be computed and independently verified in time polynomial in the triangulation size, coordinate bit length, and size of the compressed output, without enumerating individual discs or parallel product regions?

This is the most consequential implementation target for the hierarchy route. A useful representation must retain oriented attaching intervals, boundary patterns, component identities, and maps back to the original manifold. Merely recording multiplicities of local normal disc types does not say how the new boundary pieces are paired. A benchmark should therefore include exponentially large parallel families with a fixed number of local types, and compare the compressed result with a literal cut on all feasible small members. The pass condition is equality of reconstructed attaching maps and certificate replay, together with output-size and bit-cost bounds.

Research question 7.4 (Certified parallelity reduction). Can all relevant product or parallelity regions be recognized and collapsed using compressed interval data, with a certificate of the resulting handle structure and inherited boundary pattern?

This would connect the implementation to the refined hierarchy's depth control. It must include exceptional or non-disk bundle bases and their vertical boundaries, rather than assuming every repeated local block is a trivial product. A concrete first milestone is an exact list of permitted local quotient operations, each with a reconstruction certificate and explicit effect on the chosen complexity measure. The benefit should be measured in compressed handle count and subsequent reset work, not only in the number of expanded tetrahedra avoided.

Research question 7.5 (Normal curves after a compressed cut). Can boundary intersection, isotopy, and normalization operations be made polynomial in all representation parameters actually produced by cutting?

Lackenby’s fast curve algorithms provide suitable starting primitives [8]. Their hypotheses and parameters must be preserved at the adapter boundary. In particular, standard-curve normalization depends on its simplification number as well as triangulation size and logarithmic weight. Controlling only binary multiplicities is therefore insufficient if the cut creates many independently paired arc pieces. The next artifact should expose that pairing count or simplification number as a measured and proved state-size parameter, with literal small-instance comparison.

Research question 7.6 (Independent geometric decision certificates). Can a successful hierarchy, crushing sequence, or meridian disk be exported from discovery and replayed by a small exact checker bound to the original diagram and its peripheral data?

The cocycle matching is a model for this boundary: the solver proposes a small certificate and a separate routine checks the claimed objective. Geometry needs richer witnesses, including the validity of the exterior, essential boundary slopes, and the admissibility of each operation. A first positive milestone is the Euler-characteristic disk test proved in Section 3. Its implementation should not mistake a rank-one manifold for the input knot exterior or use a disk with inessential boundary as an unknot certificate.

7.5 Priority II: a global descent or reset theorem

Research question 7.7 (Reset accounting stronger than lexicographic termination). Is there an amortized potential, persistent-prefix reuse theorem, or canonical-state bound that replaces $L(g+1)^L$ by a quasi-polynomial bound on actual reconstruction work?

An informative experiment must record why a prefix is discarded, what data remain reusable, and how much later work is repeated. Counting only successful decompositions hides the potentially dominant cost. A candidate theorem should specify an injection from reset events into a bounded family of combinatorial objects or prove a multiplicative decrease after a bounded batch of events. If neither can be shown, an explicit family with repeated near-identical resets would be a valuable negative result.

Research question 7.8 (State size under repeated cutting and simplification). Can the complete compressed state remain quasi-polynomial in the original diagram size throughout every necessary branch?

The word “complete” includes coefficient bits, interval endpoints, pairings, boundary-pattern labels, and certificate history. A fixed local alphabet of handle types is helpful but does not bound the number of copies or distinct attaching patterns. A proposed invariant should have a bound under each primitive and under composition; an individual output-sensitive algorithm alone does not prevent its outputs from growing too quickly. Recording both peak size and cumulative newly created data will distinguish reusable DAG sharing from repeated copying.

Research question 7.9 (Algebraic summaries beyond the Jones state space). Can a complete detecting invariant or geometric boundary condition be represented at polylogarithmic interfaces with quasi-polynomial total algebraic multiplicity and update cost?

Replacing Bell partitions by binary states solves a specific invariant calculation. It cannot simply be transferred to Khovanov complexes: the number of boundary labels does not bound chain-group multiplicities, differential data, or the work needed to perform exact cancellation and rank decisions. Even with one boundary label, an abstract chain object can carry arbitrarily many summands. A viable stronger summary needs an exact composition theorem controlling those data, with proof that the final decision is complete. Small widths on special diagram classes should be studied separately from claims about arbitrary planar diagrams.

7.6 Priority III: improving the new primitives

Research question 7.10 (Weighted and class-aware cocycle optimization). Can the certified flow objective be extended to local geometric costs that predict hierarchy work, and can a bounded set of cohomology classes be searched without losing a global bit bound?

Positive integral tetrahedron weights preserve the network formulation, but require a flow algorithm polynomial in binary capacities. Minimizing genus or pattern-complexity is not obtained simply by renaming the disc-count objective. Varying the cohomology class also leaves the current potential family and may reintroduce an integer search problem. A useful first result would state exactly which weighted local objectives remain convex network spans, and compare their certified optima with measured cut or reset costs on genuine knot exteriors. The rank-one primitive case is a natural initial target because its class choices are controlled.

Research question 7.11 (When does additional vertex freedom justify its cost?). Can truncation or subdivision be chosen by a certified benefit/cost rule that improves seed construction or downstream geometry on difficult inputs?

The finite-exterior examples show useful freedom with ten vertices and none after one-vertex simplification. More vertices do not automatically help: they increase the flow and geometric state, and disc count changes with the triangulation. A fair comparison should report the number of tetrahedra, bits, normal discs, Euler characteristic, and actual downstream work. For known pure gauge changes on a fixed triangulation, the certificate-transport proposition avoids solving the flow again. Retriangulation requires a separate transport theorem and cannot use that formula without proof.

Research question 7.12 (Smaller tensor work with the same certified bound). Can order selection, outer-face selection, and a bounded faithful prelude reduce actual state and arithmetic costs while preserving the universal square-root bound?

The existing certificate allows heuristic orders to compete with a guaranteed one. A cost model incorporating actual state multiplicity and mantissa bits may predict runtime better than frontier width alone. Any order-search budget must be charged. An especially useful negative deduction follows from Proposition 2.6: changing only a same-outer bend gauge cannot reduce normalized mantissas or their bit lengths for a fixed order. Optimizing that gauge solely to shrink integer precision would be wasted effort. Changing the outer face is a different operation and can be tested explicitly, while retaining the complete turning proof. Sharper bounds on formal coefficient span also remain a mathematical question, rather than an assumed consequence of gauge covariance.

Research question 7.13 (Witness quality and the next compressed bottleneck). Can a budgeted overlap scheduler obtain useful gain guarantees, while rejecting impossible whole-donor queries before they consume the allowance?

The present first-witness gain ratio tends to zero on a proved family, so an unconditional approximation theorem for this exact policy is unavailable. A modified policy might retain several cheap witnesses, increase a threshold geometrically, or optimize gain only within a controlled collection of periodic alignments. Each proposal needs a clear extra-work bound and a counterexample search. Independently, the pair $(ab)^N, (aabb)^N$ calls for a cheap exact incompatibility summary of run lengths or periodic defects at the earlier whole-donor stage. The relevant parameter may be defect count in addition to grammar size and length bit size.

Research question 7.14 (From local group moves to complete bounded discovery). For presentations arising from actual knot diagrams, can one prove a quasi-polynomial bound on necessary moves, grammar growth, and certificate size for a complete discovery strategy?

This is substantially harder than a polynomial local word theorem. A strict decrease of expanded relator length may still allow exponentially many unit decreases. Strong compression does not itself make the greedy move set complete. Research should distinguish arbitrary synthetic presentations from residuals reachable by verified transformations of Wirtinger presentations. The next benchmark target is a collection of actual knot residuals above the explicit bridge cap that genuinely exercise the new partial-overlap stage, with independently replayed final certificates and all preceding-stage work included.

7.7 Milestones and decision criteria

The priorities above can be organized into three reviewable releases. The first should implement a capped full-Jones portfolio, seed-certificate transport, and a peripheral-data-bound positive disk checker. Success means unchanged exact outputs, independent certificate replay, and a documented improvement on inputs that exercise the changed path.

The second should deliver compressed cutting and normal-curve adapters with explicit output-size bounds, including product-region and boundary pairing data. Success means agreement with literal topology on small cases and polynomial dependence on binary weights on large structured families, with no hidden disc expansion. If some family forces excessive pairing growth, that obstruction should be stated before extending the hierarchy code.

The third should address (11) and (10): prove an amortized reset or canonical-state bound for a complete strategy, or construct a rigorous family showing that the proposed strategy cannot satisfy it. A general quasi-polynomial claim should be made only when the completeness theorem, state-size invariant, primitive bit bounds, and total-work accounting all refer to the same implemented or fully specified algorithm.

The work in this package makes the next stage concrete. It provides exact primitives whose guarantees can be composed when the missing global structure is available, and it identifies several attractive shortcuts that fail under direct mathematical or experimental scrutiny.

A Reproducibility and repository integration

A.1 Package map

The accompanying archive is organized as follows. Relative paths are from the archive's top-level directory.

Path	Purpose
<code>article/unknot_certified_primitives.tex</code>	Main editable article; section inputs, bibliography, generated tables, and vector figure are beside it.
<code>article/unknot_certified_primitives.pdf</code>	Compiled and visually checked article.
<code>article/build.sh</code>	Rebuilds the article with PDFLaTeX through latexmk.
<code>integration/changes.patch</code>	Repository-relative source patch against the pinned baseline.
<code>integration/tree/</code>	A runnable source snapshot preserving Topology/UnknotRecognition/ and the required reference packages.
<code>integration/manifest.json</code>	SHA-256 hashes, changed-file inventory, and baseline identity.

Path	Purpose
integration/patch-validation.json	Temporary-index apply check and byte-for-byte reconstruction of every changed file.
results/	Full raw exact outputs, timing samples, table values, completion status, and independent audits.
results/checkpoints/	Earlier tensor arithmetic measurements, explicitly distinguished from the final six-arm comparison.
certificates/	Independently replayable original-diagram Gordian certificate used by the overlap audit.
reproduce/	Test runner, independent Regina audit, table/figure generator, assembly helper, and detailed reproduction instructions.
validation/	Unedited test outputs, final coverage composition, worker-regression diagnosis, independent reviews, and article build/QA record.
provenance/	Mathematical audit notes and primary-source metadata.

The source snapshot contains the maintained `fast` code, tests, fixtures, and reference packages from reports 02, 24, 26, and 28. It does not duplicate the entire historical repository or bundle the external Regina binary. Relevant existing licenses and package metadata are retained. The final source patch modifies or adds eighteen files. It is accompanied by a full snapshot so that mathematical review is not dependent on reading a diff alone.

A.2 Applying and checking the patch

At the specified baseline checkout, first inspect the change summary and perform the ordinary Git apply check:

```
git rev-parse HEAD
git apply --check /path/to/package/integration/changes.patch
git apply /path/to/package/integration/changes.patch
```

The recorded baseline is

8a95834940cf77cdab1b39571ffc102ca8b6bede

If integrating after later repository work, review any conflicts against that baseline instead of assuming the patch’s context still matches. The delivered package was checked in a separate temporary index: the patch applied and every reconstructed changed file matched the snapshot by hash. This did not modify the checkout’s real index or create a remote commit.

The article can be placed with the repository’s research reports and linked from the synthesis documents during integration. It is self-contained and does not require historical article fragments to compile. Its proofs make clear which algorithms are newly implemented, which bounds were already known, and which routes remain conditional.

A.3 Tests and independent audits

From the source snapshot or integrated checkout’s `fast` directory:

```
python -B -m unittest discover -s tests -v
python -B audit_tensor_jones.py \
    --output results/tensor_jones_audit_local.json
python -B benchmark_tensor_jones.py --rounds 7 --seconds 2 \
    --output results/tensor_jones_local.json
python -B benchmark_cocycle_seed.py --regina \
    --output results/cocycle_seed_local.json
python -B benchmark_overlap_witness.py \
    --output results/overlap_witness_local.json
```

The optional Regina version used here is 7.4.1. Without it, the base modules remain usable but independent geometric tests are not the same validation run. The module-by-module test runner documents an alternative complete inventory route and the exact final coverage composition.

The overlap historical comparison reads the archived helper from Git at the pinned revision. It therefore requires that Git object, normally supplied by the target repository checkout. The standalone current-code snapshot suffices for its tests and ordinary use, but does not recreate absent Git history. The README records this prerequisite explicitly.

The independent Regina audit runs from the package directory:

```
python -B reproduce/audit_regina_jones.py \
    --fast-dir /path/to/ProveIt/Topology/UnknotRecognition/fast \
    --output results/regina_jones_local.json
```

An optional `-regina-path` supplies a separate dependency installation. The audit validates variable conventions and compares whole coefficient maps. It does not use a polynomial digest as a substitute for equality.

A.4 Rebuilding the article and tables

The main document uses standard TeX Live packages and no shell-escape extension. Tables and the figure can be regenerated from archived JSON:

```
python -B reproduce/make_research_tables.py
sh article/build.sh
```

The figure generator uses Matplotlib; `-tables-only` avoids that dependency when the supplied vector figure is retained. Table generation checks the median calculations against individual samples and does not rerun a benchmark. Earlier checkpoint timings are not mixed into the final table.

The PDF has been rendered to page images and inspected for clipping, unreadable tables, malformed mathematics, and unintended blank pages. Build diagnostics and the page-level review record are retained under `validation/`. Auxiliary TeX build products are unnecessary for source reproduction, apart from the supplied bibliography source and generated figure assets.

A.5 Limitations of the research record

This is a research continuation prepared with AI assistance, with separate mathematical, algorithmic, and literature audit passes. Those checks are not external peer review, and the mathematical proofs have not been formalized in a proof assistant. The article supplies enough definitions, proofs, counterexamples, and exact artifacts for further scrutiny rather than asking the reader to accept a performance claim from an informal summary.

The initial incomplete test process and the reproduced pre-fix worker failure are retained separately from successful final coverage. Timings are from a single recorded environment. The current knot

corpus does not exercise the new compressed partial-overlap path during ordinary complete queries. The normal-seed objective does not certify incompressibility, and the tensor bound concerns full Jones computation. A general quasi-polynomial bound for the maintained complete recognizer remains a goal of the research program, not a deliverable claimed by this package.

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